STATISTICS · UNIT 2: SAMPLING AND EXPERIMENTATION

The Study Music Experiment

Design a survey and a randomized experiment to find out whether music helps students study.

Honors

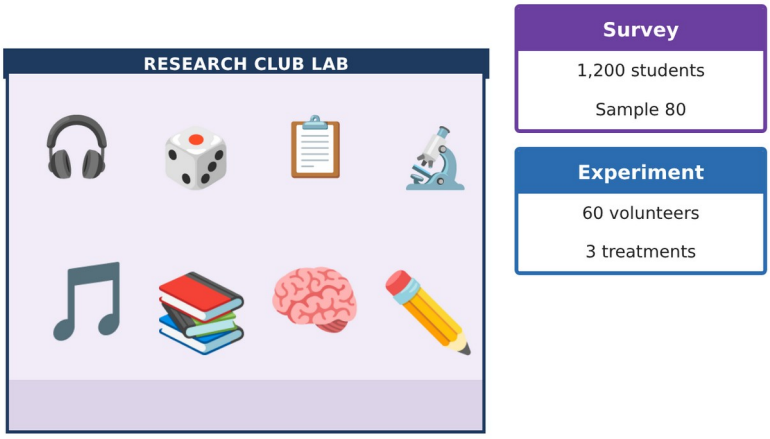
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Many students study with music on. Some teachers think it hurts focus; students say it helps. The school's research club wants a real answer, not opinions.

Your team will first survey the school about study habits, then design an experiment that tests whether music actually changes how many vocabulary words students remember.



Learning Goals

- I can choose and justify a sampling design and anticipate bias.
- I can design completely randomized, randomized block, and matched pairs experiments.
- I can use a randomization simulation and state the scope of inference.

Project Timeline

When	What you will do
Day 1	Survey Design
Day 2	Better Experiments
Day 3	Is the Difference Real?
Final	Study Design Proposal + reflection

Part 1: Survey Design

Day 1 · Use the Enrollment, Homerooms, and Random digits cards.

Research Fact Cards

Enrollment	Homerooms	Random digits
Gr 9: 360 Gr 10: 300	48 homerooms	19223 95034 05756
Gr 11: 300 Gr 12: 240	of 25 students each	28713 96409 12531

Volunteers	Treatments	Recall test
60 students	Silence · Instrumental	Study 30 words for 10 min
labeled 01–60	Music with lyrics	Score = words recalled

1. Compare an SRS of 80, a stratified sample by grade, and a cluster sample of homerooms. Which would you choose and why?

2. Use the random digits (all of 19223 95034 05756 28713 96409 12531) to choose 10 volunteers from 01–60. List them.

3. Write a neutral survey question and describe two ways nonresponse or response bias could still affect results and how to reduce them.

Part 2: Better Experiments

Day 2 · Recall ability varies a lot from student to student.

4. Design a randomized block experiment that blocks by grade level. Why block?

5. Design a matched pairs experiment comparing silence and lyrics using each student as their own pair.

6. Why must the order be randomized and the two word lists be equally hard?

7. Is a placebo possible here? Describe what could serve as one.

Part 3: Is the Difference Real?

Day 3 · Completely randomized results: silence mean 18.2 (20 students), lyrics mean 15.6 (20 students).

8. State the question the simulation answers, in terms of chance.

9. In 200 re-randomizations, 3 differences were 2.6 or more. Estimate the p-value and draw a conclusion.

10. Write the scope of inference: cause and effect? Generalize to whom?

11. Suggest one change that would let the club generalize to the whole school.

Final Product: Study Design Proposal

Your final product must include:

- Justified sampling plan with random selection
- Bias-reduction plan
- Block and matched pairs designs
- Simulation p-value and conclusion
- Proposal with scope of inference

Reflection

R1. When is a matched pairs design better than a completely randomized design?

R2. Why can a significant result still be limited in who it applies to?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Sampling and experiment designs use randomization correctly and address bias and confounding. Goes beyond the minimum with a creative, realistic choice.	Sampling and experiment designs use randomization correctly and address bias and confounding.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Proposal includes a clear diagram, a bias plan, and a correct scope of inference. Polished, organized, and visually engaging.	Proposal includes a clear diagram, a bias plan, and a correct scope of inference. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.